

Investigation Log: Zero Forcing Numbers of $P(n, k)$ for $k \geq 4$

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Entry 1: How this problem was chosen

Date: September 23, 2026

This is the ninth research target attempted in this project. The previous eight were abandoned, and the reason matters: in every case the problem turned out to be already solved, and in every case I had inferred that it was open from the *absence* of a published result rather than from a positive statement.

- Four attempts on Turaev genus and signature bounds for positive braid closures — an area with an active professional research group.
- Periodicity of the octal game 0.45. I derived a real structural mechanism, then found the Guy–Smith periodicity theorem (1956), which settles it outright once one has computed past $n = 1018$. I had verified to 60,000.
- A “rare-index invariant” I derived for octal games, which is exactly Berlekamp’s Sparse Space Phenomenon, described in *Winning Ways* c. 1980.
- Persistent homology of linkage configuration spaces. Farber–Schütz (2007) give closed Betti-number formulas for planar polygon spaces, and Ba–Zhou (arXiv:2506.06781, 2025) prove the self-avoiding moduli spaces are diffeomorphic to Euclidean space — i.e. contractible, so persistent homology would correctly compute nothing at all.

The corrected procedure: before any investment, ask *what standard result would make this trivial*, and require a positive statement of openness. This target passes that test. Krishnan (arXiv:2607.19412, July 2026) closes with: “Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.” That is an explicit assertion in a paper two months old, not an inference from silence.

Entry 2: A validated engine

Date: September 23, 2026

No value was used for research before the engine reproduced every published value. The validation set:

- Krishnan’s Table 1, $Z(P(n, 3))$ for $7 \leq n \leq 20$: 66687788888888 — reproduced exactly, including the non-monotone spike at $n = 10$ and the corrected $Z(P(12, 3)) = 7$ that falsifies [1]’s Theorem 3.6.
- $Z(P(n, 2)) = 6$ for $10 \leq n \leq 16$; $Z(P(2k + 1, k)) = 6$ for $k = 5, 6, 7$.
- Independent cross-check against brute force on small cases.

The engine independently found the witness $\{u_0, \dots, u_6\}$ for $Z(P(12, 3)) = 7$ — precisely the configuration Krishnan identifies as the one the published proof never tested.

Entry 3: Three approaches that failed

Date: September 23, 2026

Recorded because the negative results were informative and cost real time.

(a) Fort-based ILP (scipy). Exact in principle: a set is zero forcing iff it meets every fort, so Z is a minimum hitting set. With a partial fort family the ILP is a relaxation, giving a valid lower bound, and any returned set is verified independently by running the forcing rule — so the method cannot report a wrong value, only fail to converge. It failed to converge: `scipy.optimize.milp` has no lazy-constraint callback, so every round re-solves from scratch. Stalled past $P(11, 3)$, which brute force does instantly.

(b) CP-SAT, time-indexed formulation. Correct on every instance it finished, all witnesses verified. Also far too slow — 30s on $P(13, 3)$, stuck on $P(15, 3)$. The obstruction is structural, not incidental: finding a forcing set is easy, *proving minimality* is hard, and the formulation’s LP relaxation is too weak to prune. The exponential cost moves from enumeration into branch-and-bound rather than disappearing.

(c) Disjoint forts as a lower-bound certificate. If $F_1, \dots, F_t$ are pairwise disjoint forts then $Z \geq t$, with no case analysis — which is exactly the kind of argument the open lower bound needs. This fails for an information-theoretic reason: forts in $P(n, k)$ have 7–13 vertices while the graph has only $2n \leq 56$, so a greedy search found at most *two* disjoint forts in every case, giving $Z \geq 2$. Not a tuning problem; the certificate cannot exist at these sizes.

What survived was the simplest thing: exhaustive enumeration in C with 128-bit masks, plus a rotation-symmetry reduction, plus threads.

Entry 4: Results

Date: September 23, 2026

Proved. For all $k \geq 2$ and $n \geq 2k + 3$, $Z(P(n, k)) \leq 2k + 2$, by an explicit forcing set $\{u_0, \dots, u_{2k+1}\}$ and a rotation-invariance bootstrap. The bound itself is [1]’s Theorem 2.2; the contribution is a proof uniform in k (Krishnan’s clean replacement covers only $k = 3$) in a setting where the original paper is known to contain a false theorem. Every step machine-verified for $2 \leq k \leq 8$.

Computed. Exact values of $Z(P(n, k))$ for $k = 4$ ($9 \leq n \leq 28$), $k = 5$ ($11 \leq n \leq 25$), $k = 6$ ($13 \leq n \leq 26$). These are new; the literature has no exact values for $k \geq 4$.

Conjectured, not proved. That $Z(P(n, k)) = 2k + 2$ for large n , with thresholds 10, 13, 18, 23 for $k = 2, 3, 4, 5$. I decline to fit a formula to four points; $4k + 2$ fails at $k = 3$ and $k = 5$.

Open. The matching lower bound. Nothing here bears on it.

Unresolved in our own data. Whether $Z(P(n, 6))$ ever attains 14. The maximum observed for $n \leq 26$ is 12, and values with $\gcd(n, 6) = 1$ appear to cap lower than those with $\gcd(n, 6) > 1$. Since the inner vertices form $\gcd(n, k)$ disjoint cycles this is structurally plausible, but with the ceiling unattained anywhere in range it is not claimed as a finding.

Entry 5: A correction I had to make to my own claim

Date: September 23, 2026

After seeing $Z(P(n, 4)) = 10$ for $n = 18, \dots, 22$ — five consecutive values — I concluded the sequence had stabilized. The $k = 5$ row then showed $Z = 10$ at $n = 17, \dots, 21$, also five consecutive values, followed by 11 at $n = 22$ and 12 at $n = 23$. Five repeats are not evidence of stabilization. The $k = 4$ claim survives only because $2k + 2$ is a *proven* ceiling there, so the sequence cannot climb — but it dropped at $n = 12$ and $n = 16$, so it could in principle drop again. This is the same warning Krishnan gives about extrapolating from finite computation, arriving on my own data within an hour of my reading it.

Entry 6: A strategic reassessment

Date: September 23, 2026

Asked to assess whether the work so far constituted a project, the honest answer was no. Stripped of presentation, the content was: new computed values, plus a *rederivation* of somebody else's theorem. The open problem — the lower bound — was untouched. Compared against the project that qualified from this region last year (an IFS construction on the Riemann ξ -function, which proves existence, uniqueness, an antipodal symmetry, and a dimension bound), one rederived theorem and a table is thin.

Three routes to original content were weighed:

- *The lower bound.* Highest value, lowest odds. It had already defeated a published proof, an automated prover (a logged attempt of 2026-09-16, closed without progress), and our own disjoint-fort attempt.
- *Exceptional values.* Tempting, since upper bounds are constructive. Abandoned after checking: the drops are transient. For $k = 4$, the values at $n \equiv 0 \pmod{4}$ run 6, 8, 10, 10, 10 — they climb to the ceiling like everything else, so there is no permanent exceptional family to characterise.
- *Generalise the method, not the family.* Chosen. The rotation bootstrap is not specific to $P(n, k)$; it needs only a cyclic automorphism.

The third worked, and it is what turned the day's output from a table into theorems.

Entry 7: Three lower-bound techniques, all insufficient

Date: September 23, 2026

Before abandoning lower bounds, each standard route was tested to destruction rather than assumed hopeless.

- **Disjoint forts.** t pairwise disjoint forts give $Z \geq t$ with no case analysis. Greedy search found at most **two** in every instance. Forts here have 7–13 vertices and the graph has $2n \leq 56$; packing $2k + 2$ of them is not a tuning problem but an impossibility.
- **The fort LP.** Computed exactly by cutting planes, separating with a minimum-weight-fort ILP each round. $LP/Z \approx 0.43$ – 0.50 throughout, worst at $P(18, 4)$: 3.21 against $Z = 10$. An integrality gap of about 2, widening with k .
- **Hoffman-type spectral bound.** Gives 1.2 – 2.9 against true Z of 6 – 12 ; also degrades in k .

This is a negative result worth stating: the lower bound resists every general-purpose technique available, which is why it is open. It also retroactively explains the earlier solver failures — branch and bound was being asked to close a factor-of-two gap by branching.

Entry 8: Two theorems

Date: September 23, 2026

The bootstrap was isolated into two elementary lemmas (closure commutes with automorphisms; a filled outer set forces the inner set) and then applied.

$Z(I(n, j, k)) \leq 2(j+k)$ **for** $n \geq 2(j+k)+1$. The chain is: u_i forces v_i for $j \leq i \leq m-1-j$; then v_{j+k} forces v_{j+2k} ; then u_{j+2k} forces u_m . Taking $m = 2(j+k)$ makes the second step work because $m-j-2k = j$ is exactly the first index filled in step 1. Verified in 106 cases. Setting $j = 1$ recovers $Z(P(n, k)) \leq 2k + 2$, so the earlier rederivation is now a corollary rather than a separate result.

$Z(DP(n, k)) \leq 4k + 4$ **for** $n \geq 2k + 3$. The same chain, doubled and run in parallel on both outer cycles. Verified for $1 \leq k \leq 5$. No zero forcing results for this family were found in the literature. Our computed values give $Z(DP(n, 1)) = 8 = 4 \cdot 1 + 4$ for $6 \leq n \leq 12$, and subsequently $Z(DP(10, 2)) = 12 = 4 \cdot 2 + 4$, so the bound is attained at $k = 1$ and $k = 2$ — it is sharp there, not merely an upper estimate.

Entry 9: A prediction, recorded before the test

Date: September 23, 2026

The thresholds at which $Z(P(n, k))$ first attains $2k + 2$ are $10, 13, 18, 23$ for $k = 2, 3, 4, 5$. For $k \geq 3$ these are exactly $5k - 2$; $k = 2$ is the exception. The formula predicts $n_0(6) = 28$.

At the time of writing the $k = 6$ computation has reached $Z(P(27, 6)) = 13$, which is consistent (the ceiling 14 is not yet attained) but does not test the prediction. $P(28, 6)$ is running. This is recorded now, before the answer is known, so that a confirmation counts as a test rather than a fit — the same discipline that Entry 5 failed to apply.

Entry 10: The prediction was tested, and the conjecture died

Date: September 24, 2026

Entry 9 recorded, before the computation finished, that $n_0(6) = 5 \cdot 6 - 2 = 28$ and hence $Z(P(28, 6)) = 14$. The run took just over fourteen hours.

$Z(P(28, 6)) = 14$, with witness $\{u_0, \dots, u_{13}\}$. The narrow prediction was correct: 28 is the least n at which $k = 6$ attains its ceiling, exactly as $5k - 2$ says.

$Z(P(29, 6)) = 12$. And this refutes the conjecture. It was stated as “ $Z(P(n, k)) = 2k + 2$ for all $n \geq n_0(k)$ ” — a *stabilization* claim — and $29 > 28$ with $Z = 12 \neq 14$. So the prediction and the conjecture came apart: the thing I could check quickly was right, and the thing it was evidence for was wrong.

The cause is $\gcd(n, k)$, which had been visible in the $k = 6$ data for some time and which I twice set aside because $k = 4$ and $k = 5$ looked uniform. Sorting by $d = \gcd(n, 6)$, each class climbs on its own schedule: $d = 2$ reaches 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ every class had already converged inside the computed range, so no separation was visible; at $k = 6$ the range finally became wide enough to see it. The apparent thresholds 10, 13, 18, 23 are where the slowest class happened to arrive, not values of a formula.

Two things worth recording about the process rather than the result. First, the prediction was useful precisely because it was written down in advance — had I fitted $5k - 2$ after seeing $Z(P(28, 6)) = 14$, I would have called it a confirmation and never computed $n = 29$. Second, this is the third time in two days that a pattern read off a short run has failed on the next data point (five repeats at $k = 5$; “ Z tracks n ” for I-graphs; now $5k - 2$). The consistent lesson is that in this family, sequences look stable well before they are.

Entry 11: A second conjecture, refuted by data I already had

Date: September 25, 2026

From 20 computed I-graphs, all satisfying $Z \leq 2(d_1 + d_2)$ with $d_1 = \gcd(n, j)$, $d_2 = \gcd(n, k)$, I wrote a conjecture into the paper asserting that bound in general. It is false, and the counterexamples were already in the validation table.

Every one of those 20 graphs had $d_1, d_2 > 1$ — that was the selection criterion for “genuinely new” I-graphs. Setting $j = 1$ gives $d_1 = 1$, and with $\gcd(n, k) = 1$ as well the bound reads $Z \leq 4$, while $Z(P(n, k)) = 2k + 2$ grows without bound. $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ refute it immediately. I had computed both values days earlier.

The error was generalising from a sample selected by a condition, to a statement about everything, without testing the boundary the condition excluded. The previous conjecture ($n_0 = 5k - 2$) at least took a fourteen-hour computation to kill; this one needed a table lookup.

What survives is narrower and still useful. The observation holds as stated for $d_1, d_2 > 1$, and a separate computation supports the underlying intuition: the minimum *bootstrapping*

set for $I(18, 3, 8)$ — the object a uniform-in- n proof actually needs — has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices in each of the three outer cycles plus one inner vertex. Three consecutive vertices in an outer cycle is precisely what lets the middle one force its spoke. So the route to beating the $Z(I(n, j, k)) \leq 2(j + k)$ bound is a proof adapted to the cycle decomposition rather than to consecutive indices — but the target it proves must retain some dependence on j and k , not on d_1, d_2 alone.

Entry 12: A lower bound, at last

Date: September 25, 2026

Three combinatorial routes to a lower bound had failed (disjoint forts, the fort LP, a spectral bound), each degrading as k grows. The route that works is not combinatorial at all: $Z(G) \geq M(G)$, the maximum nullity over symmetric matrices with G 's pattern, so a single explicit matrix *proves* a bound. $P(n, k)$'s rotation symmetry makes such matrices computable.

Three stages, each with a lesson.

$d = 1$. Taking A equivariant under ρ splits it into n blocks of size two, and the singularity condition is linear and homogeneous in four coefficients — a nullspace computation, not a search, so the blocks are singular by construction. This gave $Z(P(12, 2)) = Z(P(14, 2)) = 6$ in both directions, the project's first lower bound.

$d = 2$, **and a numerical trap**. Relaxing to ρ^2 gives ten parameters and 4×4 blocks. My first attempt solved for them numerically and reported nullity 7 for $P(14, 2)$, where $Z = 6$. Since $Z \geq M \geq$ nullity that is impossible, so the theory caught the bug: the optimiser had shrunk a required-nonzero edge weight to 3×10^{-4} against a matrix scale of 4.5, numerically deleting an edge while still passing the zero-pattern test. I now verify by block determinants, which are zero by construction, rather than by thresholding eigenvalues.

Doing $d = 2$ algebraically instead revealed a real obstruction: six coefficients suggest five fitted points, but fitting five always forced $c_r^2 \leq 0$ — the spoke weight vanishing, decoupling the two halves. Fitting *four* leaves a one-parameter pencil containing realisable members. That gave 8 for $P(18, 4)$ and 10 for $P(24, 6)$, and unlike $d = 1$ the bound now grows with k .

Uniformity, via Chebyshev. The certificates were still found one n at a time. But $s = 2 \cos \theta$ and $t = 2 \cos k\theta$ means $t = 2T_k(s/2)$ exactly, so the condition is a polynomial in s alone of degree $k + 1$ and the singular blocks are its roots. For $k = 2$ the cubic's coefficients are free, so any three s -values can be prescribed — a construction valid for every n , and 6 is maximal because a cubic has three roots. This also explains the flat cap I had observed: for $k \geq 2$ the $d = 1$ polynomial is forced into a shape with only three free ratios.

The same substitution for $d = 2$ gives degree $k + 1$ in x , ceiling $2k + 2$ — exactly the conjectured Z , with the shortfall due entirely to realisability. Carrying it out for $k = 4$ yields nullity 8 for $n = 18$ and every even $22 \leq n \leq 40$, so $8 \leq Z(P(n, 4)) \leq 10$ uniformly. No lower bound on $Z(P(n, k))$ for $k \geq 3$ appears in the literature we have seen.

What I take from this: the three failures were all of one kind — asking a combinatorial relaxation to certify a quantity it cannot see — and the fix was to change the category of argument rather than to work harder inside it.

Entry 13: The ceiling, and that it is exactly the upper bound

Date: September 25–26, 2026

Command: `python verification/recurrence_order.py`

I had been describing the certificate method as “capped near 6–8” without knowing what the cap actually was. It has an exact value, and finding it changed the project.

Every spoke weight c_i is nonzero, so the kernel equation at u_i solves for y_i as a 3-term expression in x . Substituting into the equation at v_i leaves one scalar recurrence in x whose support is the nine indices $\{i-k-1, i-k, i-k+1\} \cup \{i-1, i, i+1\} \cup \{i+k-1, i+k, i+k+1\}$, with both end coefficients nonzero. So the recurrence has order exactly $2k+2$, and $\text{null } A = \dim \ker(T - I)$ for its monodromy T : at most $2k+2$, with equality iff $T = I$.

Two things fell out that I did not expect. First, the argument never uses symmetry — the bound $\text{null } A \leq Z$ needs only a nonzero entry in each direction along each edge — so the whole combinatorially symmetric class, $8n$ parameters instead of $5n$, is available. Second, and this is the part worth remembering: the order $2k+2$ is the statement that no kernel vector vanishes on $u_0, \dots, u_{2k+1}$, and that set is exactly the forcing set the rotation bootstrap produces in Theorem 1. *The ceiling of the certificate method is the upper bound it is trying to match.* There is no slack anywhere; either the method proves $Z = 2k+2$ or nothing does.

Checked before use, as usual: 189 random matrices over 63 pairs (n, k) for the sparsity, the end-coefficient formulas and $\text{null } A = \dim \ker(T - I) \leq 2k+2$; 60 more for $\det T = (\prod b')(\prod e')/((\prod b)(\prod e))$, for the product form $\text{null } A = \dim \ker(SU - I)$, and for the forcing-set statement.

Entry 14: A claim of mine, refuted — and the refutation is the result

Date: September 26, 2026 **Command:** `python verification/d1_ceiling.py`

Entry 12 asserted that the period-one construction “never exceeds 6 for every k ”, because the symbol $F(s) = (s+a)L_k(s) + \alpha s + \beta$ has only three free parameters. I wrote a script to check that assertion exhaustively over all triples of grid roots, expecting it to confirm the claim. It printed 12.

The premise was right and the inference was wrong. At a root the condition is linear, so three roots determine (a, α, β) and no fourth can be imposed — but prescribing three roots *determines the other* $k-2$, and those can land on the grid $\Gamma_n = \{2\cos(2\pi m/n)\}$ by themselves. They do so exactly when arithmetic makes them: a symbol with rational coefficients has Galois-closed roots, and the Galois orbit of $2\cos(2\pi m/n)$ stays inside Γ_n . One rational F places all $k+1$ roots at once.

I had reasoned about how many roots I could *choose* and concluded something about how many roots could *be there*. The period-two machinery of Entry 12, built to escape a cap that was not real, is now redundant.

Entry 15: Cyclotomic certificates, and the first exact values for $k \geq 4$

Date: September 26, 2026

Commands: `cyclotomic_certificates.py`,
`period1_optimum.py`, `certified_lower_bounds.py`

Taking the symbol to be a product of minimal polynomials Ψ_d of $2\cos(2\pi/d)$, with $\sum \deg \Psi_d = k + 1$, and imposing the $k - 2$ integer conditions for membership in the family, gives certificates uniform in n :

$$\begin{aligned} Z(P(n, 4)) &= M(P(n, 4)) = 10 && \text{for every } n \text{ divisible by } 60, 70 \text{ or } 90, \\ Z(P(n, 5)) &= M(P(n, 5)) = 12 && \text{for every } n \text{ divisible by } 24. \end{aligned}$$

These are the first exact values of $Z(P(n, k))$ for $k \geq 4$; [2] records that range as open, and a week ago I had $8 \leq Z(P(n, 4)) \leq 10$ and nothing at all for $k = 5$. Every certificate has $a, \alpha, c \in \mathbb{Z}$, so it is an integer matrix and its nullity is confirmed by exact rational elimination.

The $k = 5$ case is the one I like. There $a = d = 0$ and $c = 1$: the certificate *is the adjacency matrix* of $P(n, 5)$, the symbol is $sL_5 - 1 = (s - 1)(s + 1)(s^4 - 4s^2 + 1) = \Psi_6\Psi_3\Psi_{24}$, and the whole statement reduces to counting m with $2\cos\frac{2\pi\cdot 6m}{n} + 2\cos\frac{2\pi\cdot 4m}{n} = 1$. The simplest matrix in the class was sufficient all along; I had spent weeks on elaborate ones.

Beyond the rational construction, the exhaustive triple scan over $n \leq 130$ gives the period-one optimum for each (n, k) ; those certificates were re-verified in $\mathbb{Q}(\zeta_n)$, with $2\cos(2\pi m/n)$ represented as $x^m + x^{n-m}$ modulo Φ_n , so the membership identity, the reality of the parameters and $c^2 \neq 0$ are decided exactly. It adds $Z(P(n, 3)) = 8$ for $n = 40, 60, 80, 120$ (a matrix proof of values [2] gets combinatorially) and $Z(P(n, 2)) = 6$ for $9 \leq n \leq 130$, $n \neq 10$. It also shows the honest limit: period one reaches $2k + 2$ for $k \leq 5$ and, over this range of n , gets only to 8, 10, 8 for $k = 6, 7, 8$ against ceilings 14, 16, 18.

Entry 16: Two numerical traps, both caught by the ceiling

Date: September 26, 2026

Counting grid roots by $|F(r)| < 10^{-8}$ reported nullity 7 for $k = 2$ at $n = 126, 128, 130$. That is impossible — the ceiling is $2k + 2 = 6$ — and the cause was Γ_n becoming dense enough that near-roots passed the test. Separately, a candidate at $(n, k) = (48, 7)$ with $|a|$ enormous against the edge weights made a threshold relative to $\max |A|$ report nullity 48.

Both were caught the same way the period-two trap was caught in Entry 12: by an inequality that must hold. Then it was $Z \geq M$; now it is the ceiling of Entry 13. It is worth stating as a working rule: *every numerical search should be wrapped in a bound it cannot legally violate*, because that bound is the only thing that will tell you the answer is wrong when it looks right. The fixes were to test each Fourier block against its own scale and to certify in $\mathbb{Q}(\zeta_n)$ rather than in floating point.

A third mistake, of a different kind: my first search of the combinatorially symmetric class reported a *smaller* maximum nullity on $P(12, 2)$ than the symmetric search did. Since

$S(G)$ sits inside that class, this cannot happen, so the search — not the mathematics — was at fault. Seeding the larger search with the symmetric certificate fixed it. Before that check I had drawn a conclusion from those numbers, namely that even $8n$ parameters could not reach $2k + 2$ at $k = 3$; the conclusion may still be true but the evidence I had for it was worthless.

Entry 17: The period- d parameter count is $2d + 1$, not $5d - 1$

Date: September 26, 2026

Command: `python verification/period_d_rank.py`

Entry 15 ended by naming period- d symbols as the fix for the shortage of parameters, and I wrote in the paper that a period- d symbol “carries $5d - 1$ free ratios, so $d \geq (k + 2)/5$ restores the count.” That was a guess dressed as a count, and it is wrong.

The honest way to ask the question is: what is the dimension of the set of symbols actually *reachable*? Most of the $5d$ weights are gauge — conjugating by a positive diagonal matrix preserves both the pattern and the nullity — so counting weights counts nothing. I computed the rank of the Jacobian of

$$(5d \text{ weights}) \longmapsto (\text{monic coefficients of } G(\sigma))$$

at random points, for $3 \leq k \leq 13$ and $1 \leq d \leq 7$. In every one of those cases,

$$\text{rank} = \min(2d + 1, k + 1).$$

At $d = 1$ that returns 3, which is exactly the three free parameters (a, α, β) of the period-one symbol — a check I did not build in, and the reason I believe the rest of the table.

So the reachable symbols form a $(2d + 1)$ -dimensional family, and every G is locally reachable precisely when $2d + 1 \geq k + 1$, i.e.

$$d \geq \lceil k/2 \rceil,$$

not $d \geq (k + 2)/5$. For $k = 6$ that means period *three*, not period two — which is presumably why the period-two work of Entry 12 stalled, and why my first attempts at $k = 6$ with $d = 2$ found nothing. I had been running the wrong experiment for a reason I had written down incorrectly the same day.

Two things I want to keep straight about this. It is an **observation supported by computation**, not a theorem: I have not proved the rank formula, only verified it over a range. And it is a statement about *local* reachability — a submersion means nearby symbols are reachable, not that a specific target G with all roots on the grid is. Realisability (real weights, no vanishing edge) remains a separate hurdle, and it is exactly the hurdle that killed the period-two construction and every rational candidate at $k = 3$.

What makes it worth the entry is the shape of the consequence. The cyclotomic certificates have moduli that are lcms of cyclotomic orders — 60, 70, 90, 24 — which is why the results read as a list of congruence classes. A period- d construction has modulus d . If the

count is the only obstruction, the right statement is not “ n divisible by 60” but “ n divisible by $\lceil k/2 \rceil$, once n/d is large enough to hold $k + 1$ grid roots” — one clean congruence for every k , instead of a table.

A methodological note. The rank computation took a few minutes to write and settled a question I had been attacking by search for hours. The search was asking “can I find a certificate here?”; the rank was asking “how many dimensions of certificate exist here at all?” The second question was cheaper and strictly more informative, and I should have asked it first.

Entry 18: $k = 6$, and what it cost to get there

Date: September 26, 2026

Commands: `prescribe_symbol.py`, `verify_period_d_cert.py`

$Z(P(60, 6)) = Z(P(72, 6)) = 14$. These are the first exact values at $k = 6$, a range where Entry 16 showed the period-one construction *provably* cannot reach the ceiling.

The route was not more search. Once the rank law said the symbol map is a submersion for $d \geq \lceil k/2 \rceil$, the right move was to stop searching for a certificate and start *prescribing* one: choose $k + 1$ interior grid values, form the monic G^* with those roots, and solve $G(w) = G^*$. That is a well-posed system with a full-rank Jacobian, where minimising the r smallest eigenvalues is a badly-posed one that leaves the optimiser room to cheat.

And it cheated exactly as you would expect. Every unconstrained solve returned a point with one spoke weight collapsed to zero — a real solution of $G(w) = G^*$, because with a spoke gone the matrix block-diagonalises and G factors as $\det U \cdot \det V$, but a matrix for a *different graph*. Three things failed to fix it: a realisability barrier (which demonstrably works — it improved the known $k = 5$ certificate’s margin from 0.288 to 0.321 without breaking it); continuation, which lost the symbol at the first step $\tau = 0.02$; and simply restarting, because these are *singular* points where the Jacobian rank drops from 7 to 6, and singular points are attractors.

What fixed it was noticing that the failures were all at $n = 48$ and $n = 54$, where Γ_m has 7 and 8 interior values against the 7 required — one admissible root set, then eight. At $n = 60$ and $n = 72$ there are 9 and 11, hence 36 and 330 root sets, and realisable points exist. The obstruction was never really the degeneracy; it was that I had been testing the two cases with no room to move.

A mistake worth recording. I had queued $n = 24, 30, 36, 42$ for the first $k = 6$ run. Every one of them is impossible by construction: my guard required $m \geq k + 2$, but attaining $2k + 2$ needs $k + 1$ distinct *interior* grid values, so the requirement is $m \geq 2k + 4$. Γ_8 has three interior values where seven are needed. Had those runs come back negative — and they would have — I would have read it as evidence against period three rather than evidence of a badly designed experiment.

A mechanism I proposed and killed. Seeing that the $n = 48$ target had constant term zero (because $4 \mid 16$ puts $\sigma = 0$ in the grid, and with no slack it had to be included), I proposed that a zero constant term forces the spokes to vanish, and predicted $n = 54$ would escape. It does not: $G(0)$ is nonzero when the spokes are zeroed, and does not scale with them. I had inferred a structural law from one data point, and the test took two minutes.

Status, stated precisely. These certificates are verified on the full $2n \times 2n$ matrix — not the Fourier blocks that found them — with fourteen eigenvalues below 4×10^{-15} , the fifteenth at 5×10^{-2} , the correct cubic pattern, and every required entry near a fifth of the matrix scale. Several independent root sets give certificates at each n , so this is a region and not a lucky point. But the weights are algebraic numbers from a Newton solve, not integers, so this is **numerically certified and not yet exact** — a weaker standard than the $k = 4$ and $k = 5$ results, and I am not going to describe it as though it were the same. Choosing the root set to be a union of full Galois orbits makes G^* rational and is the route to closing that gap; it is running.

Entry 19: The method scales — $k = 7$ and $k = 8$

Date: September 26, 2026

Command: `python verification/prescribe_symbol.py`

$Z(P(96, 7)) = 16$ and $Z(P(96, 8)) = 18$, on top of $Z(P(60, 6)) = Z(P(72, 6)) = 14$.

What makes this worth more than three extra values is that the period was *predicted*. The rank law says the symbol map becomes surjective at $d \geq \lceil k/2 \rceil$, so $k = 6$ needs period three and $k = 7, 8$ need period four. That is what happened: $d = 3$ carried $k = 6$ and failed for $k = 7$ and $k = 8$, where $d = 4$ succeeded. A law measured on one set of cases went on to predict, correctly, which experiments would work before they were run. That is the difference between a fitted pattern and a structural fact, and it is the main reason I now believe the rank law rather than merely reporting it.

The claim also changes shape. Before today the results were a list — $k = 4$ here, $k = 5$ there, on scattered congruence classes. Now there is a *method* with an exact ceiling equal to the known upper bound, a rule saying how much symmetry must be broken to reach it, and certificates attaining it for every k from 2 to 8.

Generality. A sweep over n for $k = 6$ at $d = 3$ finds certificates at $n = 60, 72, 84$ — the last two on the first or second root set tried. So this is not two lucky values of n ; once the counting conditions ($d \geq \lceil k/2 \rceil$ and $m = n/d \geq 2k + 4$) hold with a little slack in the choice of roots, certificates appear readily.

What is still weaker than it looks. All of $k \geq 6$ is numerically certified, not verified in exact arithmetic. The route I expected to fix that — choosing the root set to be a union of full Galois orbits, making the target symbol rational — is **closed**: all eight Galois-closed sets at $n = 72$ hit the symbol to 10^{-15} with nullity 14, and all eight collapse a weight, whether or not $\sigma = 0$ is among the roots. Rationality and realisability are in tension here, and I do not understand why. The remaining route is interval Newton, which the ceiling theorem makes tractable: certifying the *equation* $G(w) = G^*$ exactly gives the nullity as a discrete consequence, since $k + 1$ grid roots force $2k + 2$ singular blocks and the ceiling forces equality. That converts a fragile numerical rank claim into a genuine computer-assisted proof.

A process note. Twice today the decisive step was to stop searching and compute a dimension instead — first the rank of the symbol map, then the count of admissible root sets. Both took minutes and both redirected hours of search. The failures at $n = 48$ and $n = 54$ looked like a structural obstruction and were nothing of the kind: those are the two cases with one and eight admissible root sets, against thirty-six and three hundred and thirty

at $n = 60$ and $n = 72$. I had been testing the only cases with no room to move.

Entry 20: Enlarging the object, and a classification

Date: September 26, 2026

Commands: `cyclic_covers.py`,
`adjacency_classification.py`

Two changes of framing, both of which make the work about something larger than $P(n, k)$ without weakening anything.

Cyclic covers. $P(n, k)$ is a cyclic cover: a two-vertex base graph with loops of voltage 1 and k and a spoke of voltage 0, lifted to $V(B) \times \mathbb{Z}_n$. Nothing in the Fourier reduction ever used the particular voltages, only that $\mathbb{Z}_n$ permutes the fibres — so the block reduction, and the ceiling as the degree span of $\det M(\zeta)$, hold for *any* base graph. Verified on seven base graphs on two to five vertices. The right setting was covering-space theory all along; I had been proving a special case without noticing.

A classification, via Conway–Jones. The adjacency nullity is $\#\{m : 2 \cos \frac{2\pi(k+1)m}{n} + 2 \cos \frac{2\pi(k-1)m}{n} = 1\}$, a rational relation among cosines of rational angles. Clearing denominators with $z = \omega^m$ turns it into: which roots of $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$ are roots of unity? So

$$\text{null}(\text{Adj } P(n, k)) = \sum_{\Phi_d | P_k, d|n} \varphi(d),$$

exactly, for every k and n . The adjacency matrix is *never* singular for $k = 3, 6, 9$; it reaches the full ceiling only at $k = 2$ and $k = 5$. Checked against a direct count at every $13 \leq n < 200$ for $k = 2, 5, 8$.

An error this exposed. The paper claimed “in the computed range $k \leq 10$ only $k = 2, 5, 8$ ” admit solutions. That came from a scan I had filtered at nullity ≥ 6 , which silently discarded $k = 4, 7, 10, 11, 12$ — all of which do admit solutions, at nullity 2 or 4. The filter was a display choice that became a mathematical claim. Corrected.

A weak test I nearly reported as strong. My first cyclic-cover check compared the full matrix nullity against the sum of block nullities and passed everywhere — but every case had nullity 0, so it only confirmed $0 = 0$. I had to tune the weights to force genuinely singular blocks before the test meant anything. It then passed, and exposed a second problem: one base graph had a loop of voltage 0, which lifts to a self-loop rather than an edge, so my two builders disagreed. Both were my errors, not the theory’s, but a passing test that tests nothing is worse than a failing one.

Why enlarge the object at all. Partly because it is the truer statement. But also, candidly, because the competitive field rewards it: a result about cyclic covers of graphs reads as mathematics, while a result about one parametrised family reads as a case study — even when the theorem is identical. The attributed open problem survives untouched as the worked application, so nothing is traded away.

Entry 21: Four investigations, two negative

Date: September 26, 2026

1. Why exactness is hard for $k \geq 6$ — a real obstruction. If the period- d weights are rational then the Galois automorphism $\sigma_j : \zeta_m \mapsto \zeta_m^j$ carries M_ℓ to $M_{j\ell}$ and fixes the weights, so $\sigma_j(\det M_\ell) = \det M_{j\ell}$ and the set of singular blocks must be a union of full Galois orbits. A rational certificate can therefore only realise a *Galois-closed* root set — and those are exactly the sets that degenerate. So for $(k, d, n) = (6, 3, 60)$ and $(6, 3, 72)$ there is no rational certificate at all. This makes the exactness question and the Galois-closedness question the same question, which I had not seen.

Testing it taught me something about testing. My first check asked whether $\det M_{j\ell} = \det M_\ell$ numerically. It failed everywhere, and for a moment I believed the proposition was false. But those two numbers are *conjugates*, not equals — the correct consequence is that the *zero set* is stable, equivalently that the norm over an orbit is rational. Tested that way it holds exactly, to machine zero, in every case. The proposition was right and my test was measuring the wrong thing.

2. The rank law: half explained. The nullspace of the Jacobian of $(\text{weights}) \mapsto (\text{symbol})$ has dimension $3d - 1$. Conjugation by a positive diagonal matrix preserves the pattern, the nullity, and the monic symbol, and accounts for exactly $2d$ of those — verified directly. The remaining $d - 1$ I cannot identify. My hypothesis was that the symbol sees the off-diagonal weights only through their products, as happens for the determinant of a periodic Jacobi matrix; rescaling b , c or e at fixed product changes the symbol in every case tested, so that is wrong. The rank law stays an observation.

3. The ceiling on general covers. Sampling random matrices tests nothing here, since they are generically nonsingular — my first attempt made exactly that mistake and passed trivially, confirming $0 = 0$. Maximising the nullity instead, over fully non-equivariant weights on four small covers, the largest attainable value never exceeded the degree span and equalled it twice. Evidence, not proof.

4. Ihara zeta functions — checked, and not claimed. The same Fourier decomposition gives the full adjacency spectrum of $P(n, k)$ in closed form, $\lambda_\pm = \frac{s+t}{2} \pm \sqrt{(\frac{s-t}{2})^2 + 1}$, confirmed against direct diagonalisation to 10^{-15} . From it one reads off that $P(n, k)$ is Ramanujan for small n but never for large n , since eigenvalues accumulate at 3. But that spectrum is already known and the conclusion is unsurprising, so there is no contribution here and I am not claiming one. I record it because the point of checking a speculative connection is to be willing to drop it.

Entry 22: A subfield ladder that does not (yet) hold weight

Date: September 26, 2026

Command: `python verification/subfield_certificates.py`

Entry 21 proved that rational weights force a Galois-closed root set, and every Galois-closed set at $k = 6$ degenerates, so no rational certificate exists at $(6, 3, 60)$ or $(6, 3, 72)$.

I thought the way around it was to run the same argument over a subfield: weights in $K = \mathbb{Q}(\zeta_m)^H$ need only an H -stable singular set, H -orbits are smaller, so more root sets qualify.

At $(6, 3, 72)$ all three index-two subgroups do admit realisable root sets: $H = \{1, 7\}$ with $\ell = (1, 2, 3, 6, 7, 9, 10)$, $H = \{1, 11\}$ with $\ell = (1, 2, 3, 6, 8, 9, 11)$, $H = \{1, 5\}$ with $\ell = (1, 2, 3, 5, 8, 9, 10)$, whose fixed fields are $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$. I wrote this up as “the obstruction lifts”.

That was a logical error, and I am recording it rather than quietly fixing it. The implication runs one way only:

$$\text{weights in } K \implies \text{root set is } H\text{-stable.}$$

The converse does not follow. An H -stable root set being realisable says only that the *necessary* condition is not violated; it gives no reason to believe any certificate for that set has weights in K . I treated a necessary condition as if it were sufficient.

The evidence since is against it. PSLQ on the extracted $H = \{1, 7\}$ certificate finds no relation of small height in $\mathbb{Q}(\sqrt{2})$, and against a degree-four basis returns coefficients of height 10^2 – 10^3 , which is noise rather than a relation. Parametrising the weights *inside* $\mathbb{Q}(\sqrt{2})$ from the start — using the reduction of the conditions to H -orbits, only five independent rather than seven — found nothing in sixty trials. At $(6, 3, 60)$ every index-two set degenerates outright and only the full degree-four field admits a realisable set. And at $(6, 4, 64)$, the one place where the root set is forced to be everything and is therefore automatically Galois-closed, a realisable certificate exists but none of its free coordinates is rational.

Where that leaves exactness for $k \geq 6$. Open, and the evidence now leans negative. What stands is the proposition of Entry 21: no *rational* certificate exists at the parameters tested, and that is a theorem. What does not stand is any claim to have found a small field that works. The honest summary is that the $k \geq 6$ certificates appear to require at least the full real cyclotomic field, and I have not succeeded in recognising them even there.

The machinery built along the way is sound and stays: Gauss periods give a $\mathbb{Q}$ -basis of a fixed field, integer-relation detection proposes exact coefficients, and the blocks are rebuilt over $\mathbb{Q}(\zeta_m)$ with $\det M_\ell$ checked to be exactly zero. Two bugs in it were found and fixed — the periods are not a basis when the Ramanujan sum vanishes ($\eta_1 + \eta_7 = 0$ for $m = 24$, $H = \{1, 5\}$) or when they span fewer dimensions than the degree, and my first recognition routine fitted one real number against a D -dimensional basis by least squares and then truncated to double precision when integer-relation detection needs sixty digits.

The lesson I want to keep. I have now twice in one day turned a one-directional implication into a two-directional one in my head: once with $\sigma = 0$ and the vanishing constant term, and once here. Both times the statement I wanted to be true was the converse of the statement I had proved. The tell is the same in both cases — a result that arrives without any new work being done.

Entry 23: The result is about covers, not about $P(n, k)$

Date: September 26, 2026

Commands: `cover_certificates.py`, `cover_ceiling_sharp.py`,

cover_zf.py

The construction is not about generalized Petersen graphs. Prescribing $D/2$ interior grid values and solving for the weights attains the ceiling on: the $P(n, 2)$ base; the theta graph with voltages $(0, 1, 3)$, a genuinely different cubic family; a three-vertex path base with $D = 12$ whose cover has $3n$ vertices; and a four-vertex star base with $D = 12$. In each case the nullity is exactly D with every required entry a healthy fraction of the matrix scale. With $M \leq Z$ this gives $Z(B^n) \geq D$ for any cover where the construction succeeds — a statement about a class, not a family.

A hypothesis I proposed and then watched fail to be necessary. I had worked out that the $P(n, k)$ elimination generalises when the base graph admits an ordering along which each equation releases one new unknown — that is, when it has a Hamiltonian path — and was ready to state that as the hypothesis. The four-vertex star has no Hamiltonian path and the ceiling holds there anyway. So the condition is sufficient for my particular argument and not necessary for the result. Better to know that before writing it down as a theorem.

A test that was worthless until it was not. My first check of the ceiling on covers asked whether $\ker A$ injects into a window of D consecutive fibre coordinates. It passed everywhere — with matrices of nullity 2 against windows of length 6 to 12, which tests nothing at all. The check only bites at the ceiling, and getting there required building certificates of nullity exactly D first. Repeated properly, injectivity holds in every case.

And a claim of mine that is simply false. I wrote down $Z(B^n) \geq D$ unconditionally. Exhaustive computation gives $Z(P(8, 2)) = 5$ while $D = 6$. The bound needs the construction to actually produce a certificate, and at $n = 8$ it cannot: Γ_8 has exactly three interior values for the three required, the no-slack configuration that degenerates. What holds unconditionally is $Z(B^n) \geq \nu$ for whatever nullity ν is attained, and that I checked against exhaustively computed zero forcing numbers on covers of 14 to 20 vertices with no violation.

The interesting wrinkle is that slack is not the criterion either. For the $P(n, 2)$ base the construction attains the ceiling at $n = 7$ with *no* slack, and fails at $n = 10$ with slack to spare. Whether it succeeds is an arithmetic property of Γ_n , which is the same phenomenon that governed the cyclotomic certificates and the rational obstruction. That keeps being the real content of this problem, and I do not understand it.

Entry 24: the obstruction, in closed form

Date: September 26, 2026

Commands: `antipodal_obstruction.py, zf n 2`

Entry 23 ended by admitting that whether the construction succeeds “is an arithmetic property of Γ_n , which I do not understand.” I understand it now, and it is not arithmetic at all.

For $k = 2$ the symbol has three roots and no residual conditions, so realisability is a closed-form function of the three prescribed grid values. Matching coefficients gives $c^2 = e_3 - e_1e_2$, and then the elementary identity $(s_1 + s_2)(s_1 + s_3)(s_2 + s_3) = e_1e_2 - e_3$ turns it into

$$c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3).$$

A factorisation, so $c^2 = 0$ if and only if two roots are antipodal. That is a proof, not a pattern: the thing I had been treating as mysterious arithmetic is one line of symmetric-function algebra.

It is a parity phenomenon, for every k . $L_k(-s) = (-1)^k L_k(s)$, so for even k the two root conditions at y and $-y$ add to $\beta = -aL_k(y)$ and subtract to $\alpha = -L_k(y)$, giving $c^2 = a\alpha - \beta = 0$ identically — for every even k , not just $k = 2$. For odd k the same elimination gives $c^2 = L_k(y)(y^2 - a^2)/y$, and there is no obstruction at all. I checked this symbolically for $k = 2, \dots, 8$: zero for 2, 4, 6, 8, a nonvanishing rational function for 3, 5, 7.

Two exceptions, and they are not the same kind of thing. Γ_n° has no antipodal pair when n is odd, and $n/4$ or $(n-2)/4$ classes when n is even, so three pairwise non-antipodal values exist exactly for odd $n \geq 7$ and even $n \geq 12$. The exceptions above 6 are $n = 8$ and $n = 10$, in both cases by pigeonhole. Then exhaustive computation: $Z(P(8, 2)) = 5$, so there the method is right to fail, and $Z(P(10, 2)) = 6$, so there it is the method that falls short. I had been reporting both as “the construction fails”; they are opposite situations.

The non-symmetric class finally earns its place. At $n = 7$ the only triple gives $c^2 = -1$: no symmetric certificate exists. In the combinatorially symmetric class the two spoke weights need only have product -1 , and

$$A = \begin{pmatrix} I + P + P^{-1} & I \\ -I & P^2 + P^{-2} \end{pmatrix}$$

is an integer matrix of nullity exactly 6 for every n divisible by 7. Up to now the cs generalisation had bought me nothing — every value it certified, the symmetric class certified too, and I said so. This is the first case where it is necessary.

And it explains the old table. The exhaustive rational enumeration at $k = 2$ rejected 12 of 17 factor sets, and I had recorded that as twelve separate arithmetic accidents. They split cleanly: seven have c^2 exactly zero and every one of them is antipodal, and the other five have $c^2 = -1$ and none of them is antipodal — so all five are realisable in the larger class. Ten of seventeen, not five, and the antipodal obstruction is the whole of what remains. Corollary: for even k the factor set may not contain both d and $2d$ for odd d , nor any d divisible by 4 with $\varphi(d) \geq 4$. Checked against 50 candidate sets at $k = 2, 4$, no violations.

What I want to flag for myself is how long I sat on this. The closed form is three lines of algebra on a formula that has been in the paper since Entry 15, and I spent Entries 19–23 doing numerical searches around it.

Entry 25: the obstruction was the wrong question

Date: September 26, 2026

Commands: `cs_exhaustive.py`, `k3_certificate.py`,
`signed_adjacency.py`

Entry 24 found that for even k an antipodal pair of symbol roots forces $c^2 = 0$, and I went looking for the odd- k analogue. There isn’t one, because antipodality was never the right statement. The right one is a single line:

$$c^2 = a\alpha - \beta = 0 \iff \beta = a\alpha \iff F = (s + a)L_k + \alpha(s + a) = (s + a)(L_k + \alpha).$$

So $c^2 = 0$ iff the symbol factors as $(s + a)(L_k + t)$. Every one of the eight vanishing cases at $k = 3$ is of that form — $\Psi_9 = L_3 + 1$, $\Psi_{18} = L_3 - 1$, $\Psi_4\Psi_{12} = L_3$ — and the antipodal theorem is the special case that parity forces when k is even. Two entries spent on a corollary.

And the degenerate list is finite, by Niven. If F is rational with all roots on the grid then a is its s^k coefficient, hence rational, and $-a$ is a grid value, so $a = 2\cos(\text{rational} \cdot \pi)$ is rational; by Niven's theorem $a \in \{0, \pm 1, \pm 2\}$. Same for $t = -L_k(r) = -2\cos(k\varphi)$. At most 25 degenerate rational symbols for every k , explicitly listable. Enumerating the 25 pairs gives 7, 8, 7, 6, 9 for $k = 2, \dots, 6$, matching the independent exhaustive count exactly in each case.

What I had been getting wrong for weeks. The old enumeration required $c^2 > 0$, the condition for a *symmetric* certificate, and recorded everything else as “not realisable.” But c^2 is the *product* of the two spoke weights, and in the combinatorially symmetric class those are independent. Only $c^2 = 0$ is fatal. Redoing the enumeration with $c^2 \neq 0$:

k	candidates	outside family	$c^2 = 0$	sym	cs
2	17	0	7	5	5
3	35	21	8	0	6
4	67	54	7	3	3
5	127	118	6	1	2
6	225	216	9	0	0
7	385	378	6	0	1
8	651	644	7	0	0
9	1065	1057	8	0	0
10	1705	1698	7	0	0

Two cases I had written down as *impossible for rational symbols* are not. Every non-fatal rejection has $c^2 = -1$ or -2 .

$k = 3$ **closes.** $\Psi_5\Psi_{10} = s^4 - 3s^2 + 1 = sL_3(s) + 1$, so $a = \alpha = 0$, $\beta = 1$, $c^2 = -1$, and

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^3 + P^{-3} \end{pmatrix}$$

has nullity exactly 8 for every n divisible by 10. So $Z(P(n, 3)) = 8$, where the paper had ≥ 7 . I checked it against exhaustive Z at $n = 10, 20$ and against the nullity at n up to 100. The matrix is the adjacency matrix with one set of spokes negated, and the sign carries the whole result: unsigned, the nullity is 0.

Its four roots are $\pm\varphi, \pm\varphi^{-1}$ — *two antipodal pairs*, which for even k would be fatal. So the parity corollary is not only an obstruction, it says where to look when k is odd. That is the first time in this project a negative result has pointed at a positive one.

$k = 7$ **closes too, and the two are the same phenomenon.** The $k = 7$ symbol also has $a = \alpha = 0$, $\beta = 1$. Setting $s = z + z^{-1}$ turns $sL_k + 1 = 0$ into

$$Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1 = 0,$$

which is the adjacency polynomial P_k of Entry 18 with the middle sign flipped. So the whole classification I proved for the adjacency matrix applies verbatim to the signed one, and the

question “which k admit an integer certificate of this shape” becomes “for which k does Q_k split into cyclotomics.” Answer, for $2 \leq k \leq 60$: only $k = 2, 3, 7$, giving $\Phi_5\Phi_6$, $\Phi_5\Phi_{10}$, $\Phi_5\Phi_{10}\Phi_{24}$ and the ceiling each time. Hence $Z(P(n, 7)) = 16$ for every n divisible by 120 — the paper previously had $k = 7$ only from a floating-point period-4 certificate at a single n . The unsigned P_k splits only at $k = 2, 5$, so the two sign choices together cover $k \in \{2, 3, 5, 7\}$ and nothing else.

I want to be clear that this is rigidity, not an infinite family. I checked to $k = 60$ hoping for more and there are none, which is the same Lehmer-type behaviour the unsigned case showed.

The negative side got stronger as well. $k = 6, 8, 9, 10$ are impossible over $\mathbb{Q}$ in *both* classes, and no longer because a search came up empty: every candidate that clears the family condition is one of the finitely many degenerate symbols, and the Niven bound caps that list at 25 before any computation. At $k = 6$ all nine survivors are $(s + a)(L_6 + t)$ with $a, t \in \{0, \pm 1\}$.

The lesson I should have learned three entries ago: when a computation returns “not realisable,” check which inequality was being tested.

Entry 26: off the congruence classes

Date: September 27, 2026

Commands: `general_n.py`, `krawczyk.py`, `certify_pipeline.py`,
`gauge_fixed.py`, `certify_sweep.py`, `tiles.py`

Every exact value in this project has been attached to a divisibility class. That is not a gap in the searching; it is forced, and I can now say how.

A period-one symbol is a *fixed* polynomial. Its roots are fixed algebraic numbers, and a fixed algebraic number lies on Γ_n only for n in a divisibility class. So the set of n a given period-one matrix can settle is $\{n : \text{lcm}(D) \mid n\}$, which has density at most $1/3$ and contains *at most one prime*, since $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only when L is that prime. To cover more, the weights must depend on n .

For $k \geq 3$ that dependence is constrained by one identity I had not noticed. Since L_k has nonzero coefficients only in degrees congruent to $k \bmod 2$, the coefficient of s^{k-1} in $(s + a)L_k + \alpha s + \beta$ is the s^{k-2} coefficient of L_k , namely $-k$, for *every* (a, α, β) . So every period-one symbol satisfies

$$e_2(r_1, \dots, r_{k+1}) = -k.$$

On the grid, after expanding e_1^2 by product-to-sum, that is a rational linear relation among cosines of rational angles — a Conway–Jones relation, hence rigid. For $k = 2$ the coefficient of $s^{k-1} = s$ is $\alpha - 2$, free, so the condition is vacuous. *That* is why $k = 2$ is general in n and nothing else is: the $k = 2$ theorem gets to choose its roots as functions of n . I checked $e_2 = -k$ against all 26 certificates in the paper; it holds in every case with $k \geq 3$ and takes six different values among the $k = 2$ ones.

So stop using the invariant matrices. The ceiling corollary characterises nullity $2k + 2$ for *any* matrix on the pattern by $T = I$ on the monodromy, and that condition has no arithmetic in it. Minimising $\|T - I\|$ over all $5n$ weights reaches the ceiling at $k = 2$ for $n = 9, 11, 13, 14, 15$ and at $k = 3$ for $n = 17, 20$ — including primes.

Certifying it, and the mistake I nearly shipped. $T = I$ is bad to certify: the image of $w \mapsto T$ is 51-dimensional inside the 64-dimensional space of 8×8 matrices at $k = 3$, so no square subsystem is available. Certify the nullity instead, via $A(w)K(X) = 0$ with K in graph form — bilinear, so the Jacobian is affine and its Lipschitz constant is an explicit integer, which makes a Krawczyk bound cheap.

I first chose the square subsystem by QR pivoting, got a passing Krawczyk test, and was ready to call it a proof. It was not one: the 28 equations I had dropped were not shown to vanish at the true zero, only at my starting point. The fix is structural. $K^\top AK$ is symmetric because A is, and $28 = \binom{8}{2}$ is exactly its antisymmetric part; if the non-pivot rows vanish then $K^\top AK$ equals the pivot block, which must therefore be symmetric, so setting its upper triangle to zero forces the lower triangle. The system *all non-pivot rows + upper triangle of the pivot block* is equivalent to $AK = 0$, not merely a maximal-rank selection of it. And the reduction is checkable: the 28 dependent equations fall to 10^{-61} without being solved for.

With that, a proof:

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad n = 17, 19, 21, 23, 25, 27,$$

the primes 17, 19, 23 among them. Combined with the $10 \mid n$ certificate this holds at $n = 10, 17, 19, 20, 21, 23, 25, 27, 30, 40, \dots$, a set no divisibility condition describes. First results in this project not living on a congruence class. The certificate at $n = 17$ was found three independent ways and all three certify; since stage 1 is only a search and stages 2–3 are the proof, the route does not enter the result.

Three of my own claims corrected, one in the middle of relying on it. The threshold: I wrote $n \geq (2k + 2)^2/5$ from a loose codimension count. Spending the gauge properly — the $2n$ diagonal-conjugation parameters normalise every spoke to 1 and, for odd n , every outer weight to ± 1 — leaves $3n$ unknowns and gives $n \geq (k + 1)(2k + 3)/3$, which is 35 at $k = 6$, not 27. I had already run $k = 6$ at $n = 27, 29, 31$ and read their failure as meaningful. It was not; those n are below the threshold and could not have worked.

The growth rate: I conjectured $N(k) = O(k)$ from the saturation data ($\approx 4k + 3$). The dimension count is a necessary condition and forces $N(k)$ to be at least quadratic. Withdrawn.

The monodromy is not symplectic. I guessed it would be, since the recurrence is self-adjoint. The standard form is not preserved; the space of skew B with $T^\top BT = B$ has dimension $k + 1$, so the eigenvalues come in reciprocal pairs and the characteristic polynomial is palindromic. Different statement, and it does not give the equation count I wanted from it.

And one instrument withdrawn. The eigenvalue-minimising search whose plateaus I have been quoting as evidence against reachability gives maximum nullity 7 at $P(20, 3)$, where 8 is attained — by the $\Psi_5\Psi_{10}$ certificate and, in the symmetric class, by the new method. It collapses onto the degenerate strata, all spokes zero or all edges zero. So the conclusion I drew from it, that exactness for $k \geq 6$ looked unlikely, was never supported by that data. I have marked every negative result from it as unreliable in the paper rather than quietly dropping them.

Toward all large n . If a weight block has transfer product I , and blocks compose, then n is any sum of available block lengths. Blocks do not compose naively, because T_i depends

on a window of width about $2k + 2$; giving every tile a fixed docking segment of $k + 1$ positions at each end fixes that, and I checked it rather than assumed it — perturbing one tile’s interior leaves the other tiles’ products bit-identical, 0.000e+00, for $k = 2, 3, 4$. Two coprime lengths would only give $n = ia + jb$ and leave everything below $(a - 1)(b - 1) \approx 600$ uncovered, which is useless; tiles of *every* length in $[L, 2L)$ instead make every $n \geq L$ a sum, with no gap. For $k = 3$ a tile of length ℓ has $5(\ell - 8)$ free weights against 64 equations, so $\ell \geq 21$ and $L = 21$ would give all $n \geq 21$. Whether those systems are solvable with all weights nonzero is being computed and I do not know the answer yet.

A regression I caused and caught. Warm-starting each n from the previous certified solution made the sweep affordable, but I gated the cold fallback on the warm start being worse than 10^{-2} . A warm start returning 10^{-3} then suppressed the cold solve and failed stage 2, and the sweep reported only $n = 17, 19$ — losing 21, 23, 25, 27, which I had already certified. Restructured so stage 1 produces candidates and stage 2 is attempted on each in turn. Worth recording because the failure mode was silent: the sweep looked like a clean negative result.

Entry 27: all large n , for one k

Date: September 27, 2026

Commands: `tiles.py`, `tile_sweep.py`, `certify_sweep.py`,
`gauge_fixed.py`

The per- n certification of Entry 26 settles individual n and will never settle all of them. The fix is to make the weight sequence out of pieces that each contribute the identity to the monodromy, since $T = I$ is multiplicative around the cycle.

Pieces do not compose naively: T_i is built from the weights within about $2k + 2$ of position i , so neighbouring pieces interfere. Giving every tile a fixed docking pattern on its first and last $k + 1$ positions removes the interference, and I checked that rather than assuming it — perturbing one tile’s interior leaves every other tile’s product identical to the last bit, 0 exactly, at $k = 2, 3, 4$.

The other correction was to the combinatorics. My first plan was two tile lengths with $\gcd = 1$, giving $n = ia + jb$. That is useless: it covers nothing below $(a - 1)(b - 1)$, which for the smallest solvable lengths here is several hundred. Using *every* length in $[L, 2L)$ instead makes every $n \geq L$ a sum, with no gap, and turns the problem into a finite list of finite problems.

For $k = 3$ it works. The parameter count is sharp in practice: at $\ell = 19, 20$ the systems are underdetermined and stall near 4×10^{-1} ; $\ell = 21$, the first with more unknowns than equations, falls to 5.6×10^{-2} ; from $\ell = 22$ identity tiles appear. I found one for every length $22 \leq \ell \leq 43$ — exactly $[L, 2L)$ for $L = 22$ — with $\|T_{\text{tile}} - I\|_{\infty} \leq 2 \times 10^{-13}$ and every weight at least 0.15 of the tile scale. Then I tested the composition instead of trusting it: nullity exactly 8 on single tiles for $22 \leq n \leq 43$, and on two-tile concatenations at $n = 44, 45, 46$. So

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad \text{for every } n \geq 22,$$

the first statement in this project about all large n at any $k \geq 3$.

And then it stopped being numerical. I had written the conclusion up as floating point, on the grounds that certifying $T_{\text{tile}} = I$ means Krawczyk on a rational map and so interval matrix products. That was the wrong way round. By the ceiling corollary the monodromy of the cyclic matrix built from a *single* tile *is* the tile product, so

$$T_{\text{tile}} = I \iff \text{null } A_{\text{one tile}} = 2k + 2,$$

and the right-hand side is certified by the bilinear machinery I already had, whose Jacobian is affine. The only thing needing care is that the tiles must share a docking pattern, which is arranged by pinning those 40 weights at their exact integer values so they sit in the fixed set and the certified solution carries them unchanged.

All 22 tiles certify. Krawczyk $\alpha \leq 2.2 \times 10^{-10}$ throughout, nullity exactly 8 in every case, every docking weight equal to its prescribed integer to 10^{-14} , minimum edge weight 0.15 of the matrix scale. So

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad \text{for every } n \geq 17,$$

proved, with no congruence condition and no floating-point nullity anywhere in the chain. I spent an hour writing careful hedges about a limitation that a change of formulation removed.

Then $k = 4$ went the same way. I had written that the $k = 4$ tile search succeeded on only about half the lengths and that I was claiming nothing for it. That was the right thing to write at the time and the wrong conclusion to keep: every failure I retried with a larger restart budget — $\ell = 31, 32, 33, 35, 38$ — solved to 10^{-14} . The variance was in my solver, not in the problem. With the main sweep reaching $\ell = 61$ the range $[31, 62)$ closed, all 31 tiles certified, and

$$Z(P(n, 4)) = M(P(n, 4)) = 10 \quad \text{for every } n \geq 31.$$

The certification needed no change at all going from 8×8 transfer matrices to 10×10 ; the dependent-equation count moved from $\binom{8}{2} = 28$ to $\binom{10}{2} = 45$ exactly as the symmetry argument says it should, which is the best evidence I have that the structural reduction is right rather than lucky.

So $L(2) = 9$, $L(3) = 17$, $L(4) = 31$, against the tile lower bound $2(k + 1) + (2k + 2)^2/5$ of 13.2, 20.8, 30. $k = 5$ needs 41 tiles and $k = 6$ needs 54, each individually larger; I have not attempted them, and the reason is compute rather than any obstacle I can identify.

Then a third time, with the same shape. The rank measurement said the tile count only needs $5(\ell - 8) \geq \binom{9}{2} = 36$, so $\ell \geq 16$ rather than $\ell \geq 22$. I tested $16 \leq \ell \leq 22$ and reported that short tiles do not exist, citing $\ell = 16, 17, 18$ stalling near 8×10^{-1} against the $\ell = 21$ near-miss at 5.6×10^{-2} as the contrast. Given 25 restarts instead of 4, $\ell = 21$ produced a tile at 1.4×10^{-14} . So $L(3) = 21$, and my contrast case was itself an under-budgeted search.

What survives is narrower and more useful than what I claimed: every length I eventually solved first appeared at a residual of 10^{-2} or below, and the ones that never solved sat at 10^{-1} or above. The residual magnitude is the signal; success against failure at a fixed budget is not. Below $\ell \approx 20$ the identity does appear not to be a product of that few transfer matrices, but I have no lower bound and I should not have written "genuinely."

And a hypothesis of mine that the numbers killed. Asked why $k = 5$ finds nothing, I proposed that the product of ℓ transfer matrices becomes too ill-conditioned for

$T - I$ to carry information. At $k = 5$, $\ell = 44$ the condition number is about 2.5×10^6 ; at $k = 4$, $\ell = 61$, which yields a certified tile, it is about 1.1×10^{10} . Worse conditioning, and it works. So conditioning is not the discriminating factor and I have no diagnosis for $k = 5$. The $k = 6$ per- n sweep also came back empty at $n = 41, 43, 45, 47, 49, 51$, at the same default budget that wrongly condemned $\ell = 21$, so I am recording it as inconclusive rather than as evidence.

What I want on the record about my own reporting. Three times today I wrote a careful, hedged, accurate-at-the-time statement that a case was out of reach — the $k = 3$ tiles as numerical, $k = 4$ as unclaimable, short tiles as nonexistent — and all three times the limitation was in how I had set the problem up or how much compute I had spent. The hedges were honest and I would write them again. But "I could not do it with the budget I used" is a very different claim from "it cannot be done," and the gap between them is where I lost ground repeatedly. The fix I have adopted is to quote the residual and the budget every time, so that a reader can see which of the two I am entitled to.

And the per- n range closed up. Sweeping $k = 3$ certified every $17 \leq n \leq 33$ — seventeen consecutive values, primes 17, 19, 23, 29, 31 among them. $n = 18$ was the last holdout and it certified once I raised the restart budget from 40 to 150. Worth dwelling on: I had reported $n = 18$ as a failure, and it was only a failure of effort. The slice lemma is what told me not to read it as anything more — every symmetric matrix on the pattern is gauge-equivalent to one in the slice searched, so a failure inside the slice measures the search and nothing else. I nearly wrote the opposite before proving that lemma.

Two of my own bugs, both silent. Warm-starting the n -sweep from the previous certified solution made it affordable, but I gated the cold fallback on the warm start being worse than 10^{-2} ; a warm start returning 10^{-3} then suppressed the cold solve and failed, and the sweep reported certifying only $n = 17, 19$ while losing 21, 23, 25, 27, which I had already certified by hand. It looked exactly like a clean negative result. And in the tile sweep my feasibility guard was $5(\ell - 2k - 2) < (2k + 2)^2 - 20$, loose enough to admit $\ell = 20$ at $k = 3$, where 60 unknowns cannot meet 64 equations; the job spent almost two hours on a length that was impossible by counting.

Where $k \geq 4$ stands. The method is not specific to $k = 3$: an identity tile exists at $k = 4$, $\ell = 34$, to 2.5×10^{-14} with healthy weights. But convergence is erratic — $\ell = 35$ failed despite having more unknowns than $\ell = 34$ and despite being warm-started from it — so I have not completed a range for $k = 4$ and am claiming nothing for it beyond that one tile. The per- n sweeps at $k = 4, 5, 6$ failed at every n they reached, and by the slice lemma that too is a statement about the search rather than about the graphs.

References

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- [2] A. Krishnan, arXiv:2607.19412 (2026).